

* Introduction:-

Electrochemical cells -

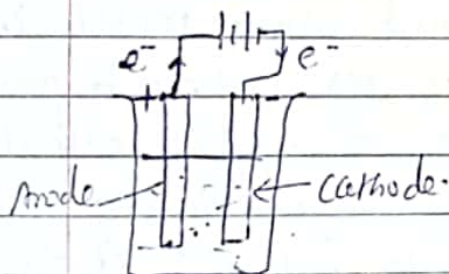
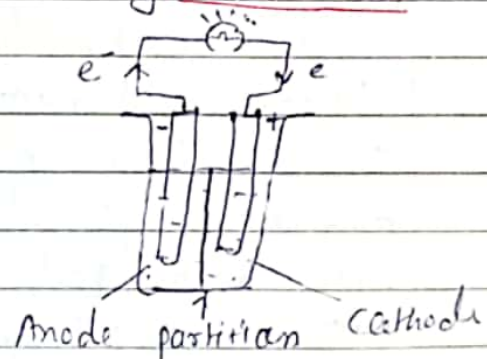
It consist of two electrodes are placed in same or diff. electrolyte which are connected by salt bridge.

Types - 1) Electrolytic2) Galvanic or voltic cell.1) Electrolytic:-

Electrolysis takes place by passing electric current which convert electrical energy into chemical energy.

2) Galvanic / voltic cell:-

chemical energy converted into electrical energy by redox reac. electricity is produce in diff. compartment by placing suitable electro-
lyte sol. & metallic conductor as electrodes. two compartment connected by salt bridge & electrodes are joined by wire thr galvanometer.

Electrolytic cellgalvanic cell

current flow

C → A
(-) (+)

sign
e- flow
Half cell?

Electrolyte cell

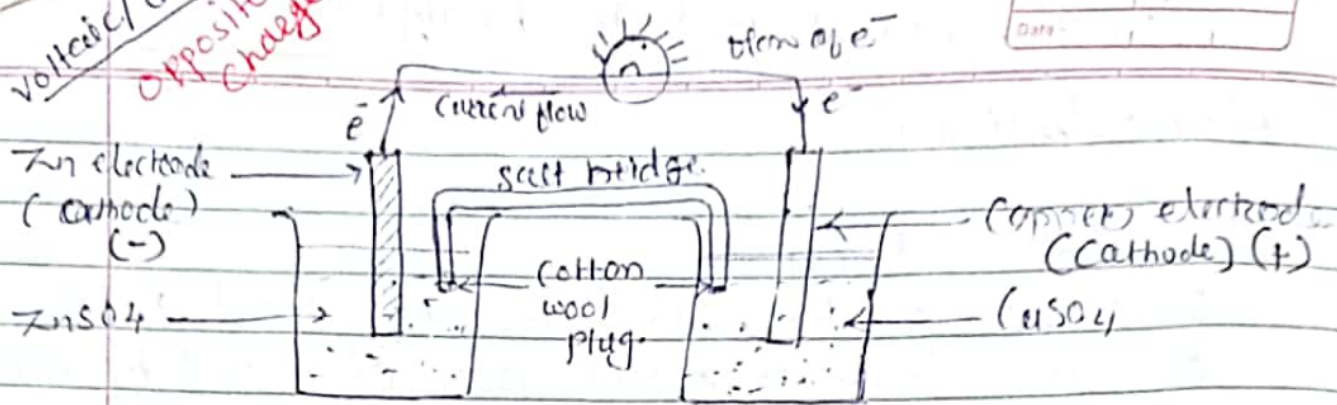
Anode Cathode

+ out
oxid.
- in
Red.

A → C Voltaic / Galvanic cell

(-) (+) Anode Cathode

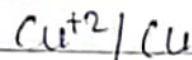
- out
oxid.
+ in
Red.



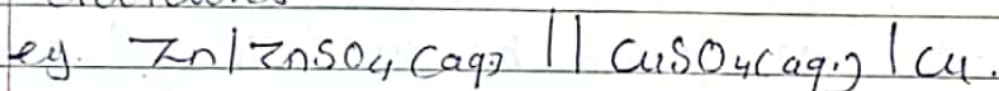
Daniell cell.

Representation of an Electrochemical cell (Galvanic)

1. Anode on left hand side cathode on right hand.
2. Vertical line (|) or (|) semicolon shows contact betⁿ two phase. 1st metal then metal ion in anode cell.
 $Zn | Zn^{2+}$ or $Zn ; Zn^{2+}$
3. molar conc. written in bracket after formula $Zn | Zn^{2+} (1M)$
4. For cathode cell - 1st cation of the electrolyte then metal.



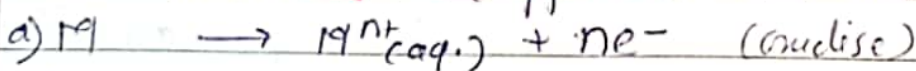
5. The salt bridge which separates the two half cells is indicated by two || or line. (vertical)
6. Sometimes +ve & -ve signs are also put on the electrodes.



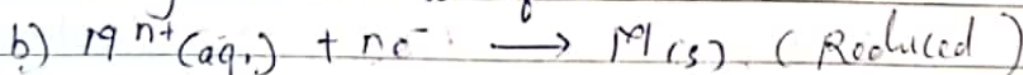
* Electrode potential:-

Electric double layer is formed when metal is placed in solution. Metal acquires either a positive or -ve charge w.r.t solution so double potential difference get developed.

- 1) When metal rod dipped in its salt solⁿ it gives



leaving an excess of e^-

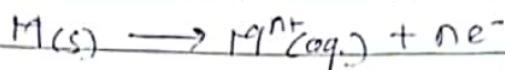
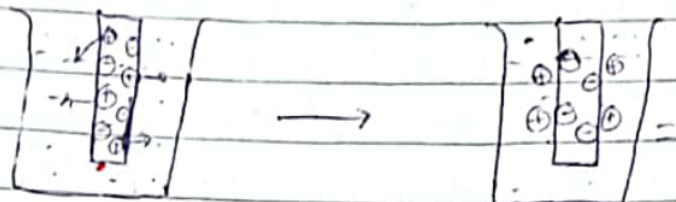


metal ions gain e^- from electrode

- 2) $M \rightleftharpoons M^{n+} + ne^-$ equilibrium is established

3. Magnitude of electrode potential of a metal is measure the loss or gain of e^- it depends on.

- 1) Nature of electrode.
- 2) Conc. of the ions in solution.



* Two types of electrode potential:-

1) Oxidⁿ potential - When electrode is -vely charged w.r.t. solution. i.e. it act as anode.

2) Redⁿ potential - When electrode is +vely charged w.r.t. solⁿ i.e. act as cathode.

* It is not possible to measure the single electrode potential directly. potential diff. can be measured. It is req^d some electrode ^{known potential} to couple with that electrode. This electrode is known as reference electrode.

$$\begin{aligned} \text{Emf of the cell} &= E_{\text{anode}} + E_{\text{cathode}} \\ &= \left(\text{Oxidⁿ potential of anode} \right) + \left(\text{Redⁿ potential of cathode} \right) \end{aligned}$$

* Types of Reference Electrodes: Well-known electrode "Electrode which has stable & reproducible potential & completes the cell action as half cell."

* Types of Reference electrodes.

- a) Primary reference electrode. eg - ^{std.} Hydrogen electrode
- b) Sec. Reference electrode - calomel electrode, Silver, Silver chloride, mercury-mercury sulphate electrode, mercury oxide electrode, quinhydrone electrode.

HE - Used for determining the cell half cell potential & reactions.
value standard electrode potential is 0

Page No.
Date

Hydrogen Electrode

* Primary Reference electrode: - (Disadvantages)

(Reproducible but) platinum used in this electrode get easily poisoned by impurities from solution. So electrode is no longer behaves reversibly.

2. It can not be used in presence of reducible ions having +ve charge electrode potential like Cu, Ag, Au

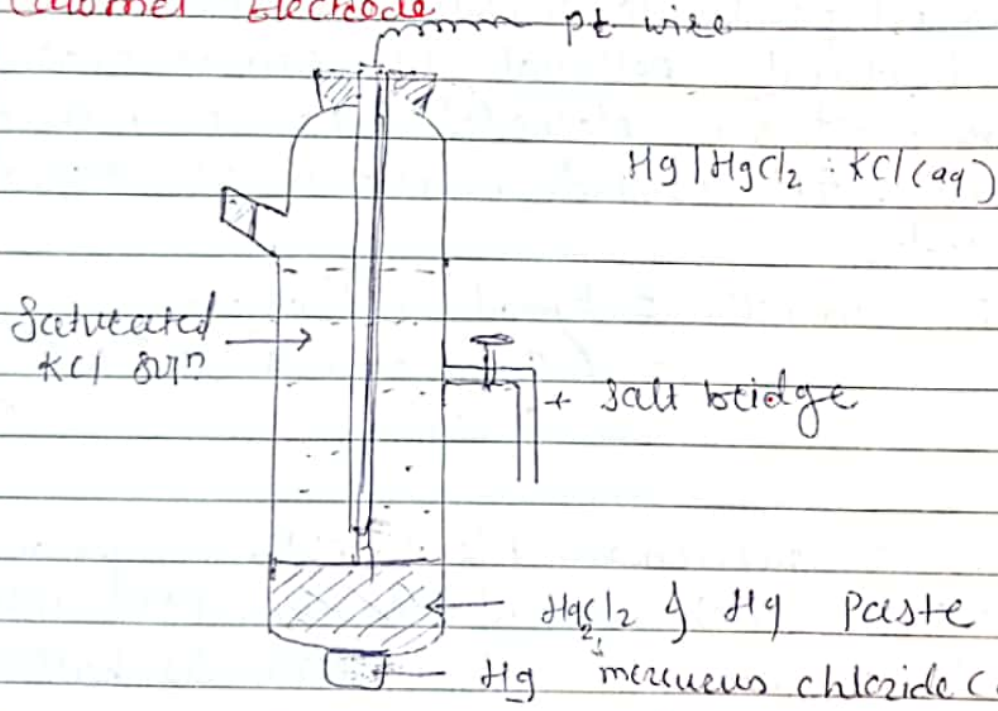
3. Presence of oxidising agents (Organic comp, alkalis, ions) in solution displace the equilibrium & change the potential.

4. difficult to prepared & maintain.

from above difficulty Secondary electrode is used.

* Secondary Reference electrode:-

Calomel Electrode



1) It consist of outer glass tube. At the bottom of this tube mercury is there. above this mercury layer a paste of mercurous chloride is placed.

2. A solution of potassium chloride is then placed over the paste.

3 platinum wire is sealed in glass tube. which help to electrical contact.

4 electrode is connected by side tube th^r a salt bridge with other electrode to make a complete cell

calomel electrode represented as -
 $\text{Hg} | \text{Hg}_2\text{Cl}_2 \cdot \text{KCl (M)}$

Reduction Conc.	Potential E°	Name
of KCl	calomel	Decinormal
0.1 N	0.3334 Volts	Normal Decinormal
1.0 N	0.2810 Volts	Normal calomel
Saturated	0.2422 Volts	Saturated calomel

If calomel acts as cathode



$$E_{\text{cal}} = E^\circ - 0.0591 \cdot \log[\text{Cl}^-]$$

If calomel act as anode:



* Demerits -

1) Not use above 70°C as Hg_2Cl_2 starts decomposing

2) involves handling of poisonous Hg & Hg_2Cl_2

The electrode potential of any electrode can be measured when it combine with calomel electrode

* Indicator electrode ∴

The electrode of a cell in which the potential depends on the conc. of a particular ion is called as indicator electrode. eg- Glass electrode

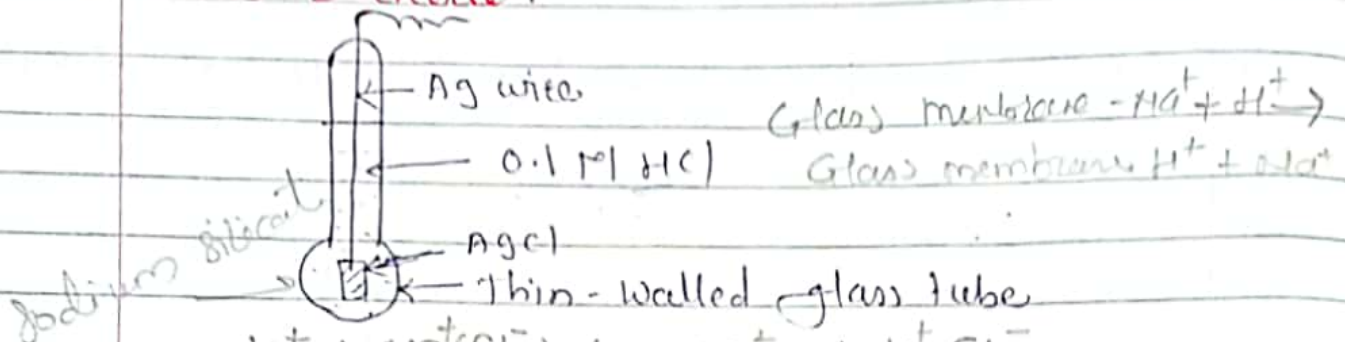
Principle ∴ when two solⁿ of different $[\text{H}^+]$ are separated by a thin glass membrane a potential diff. developed is proportional to the

diff. in $[H^+]$ of the two solutions.

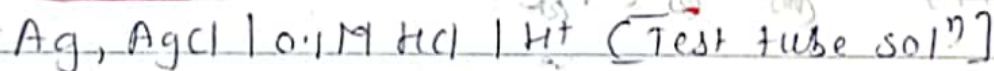
This observation was made by Haber & Klemensiewicz.

- * Glass electrode can be used almost in all the solutions, except strong acidic or alkaline.

* Glass Electrode :-



- Glass electrode is made up of very thin walled glass bulb, with low melting point glass, having high electrical conductivity. ^{blown} at the end of the glass tube.
- The bulb contains 0.1 M HCl solution. One silver wire is dipped inside the solution with silver chloride at its lower end, forming silver-silver chloride electrode.



- Glass electrode req^d soaking in water for some hr. before use as that -

- 1) Surface becomes active.
- 2) Hydrated layer get developed on outer surface.
- 3) Ion exchange takes place.

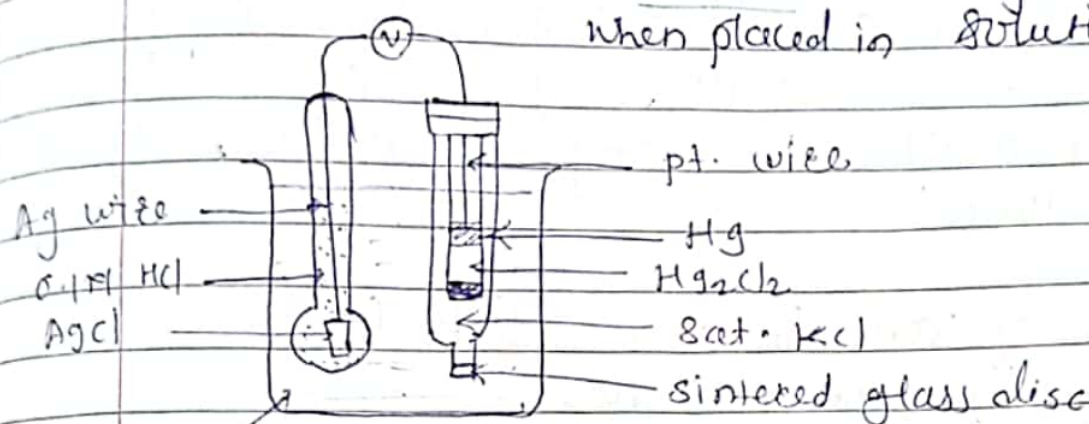
* Advantages :-

- 1) It is stable electrode used in presence of strong oxidizing & reducing agents.
- 2) It is portable & compact.
- 3) Equilibrium is easily attained.

4 It can be detect & estimate H^+ ion in presence of other types of ions.

5) Simple to use.

Potential of glass electrode measure with using calomel electrode when placed in solution (HCl)



solⁿ of unknown pH

A glass electrode is coupled with the calomel electrode to determine the pH of the solution.

Ion-selective Electrode (ISE) / specific selective electrode

The electrodes that determine the activity of ions in aqueous solution by measuring the electrical potential are called ion selective electrode.

Q ISE can measure the conc. of certain ion in solution & not affected other ions?

→ This is done by use of selective membrane which allow only specific ions to sense.

* In this electrode non porous membrane separate the two diff. conc. of same ions. So potential is developed.

Solution 1	Membrane	Solution 2
E_1		E_2

* Advantages: -

- 1) It is relatively cheap & easy to operate.
- 2) They are resistant to chemical attack.
- 3) It has wide range of conc. measurement.
- 4) It is useful in biological / medical appl. it can be used for coloured or turbid solutions.
- 5) It can determine both +ve & -ve charged ions.
- 6) It can monitor the change of activity of ion with time.

* Types of ion selective electrodes

- 1) Glass membrane electrodes: -
- electrodes are H^+ sensitive. [H^+ ions exchange with Li^+]
- 2) Solid State membrane electrodes (crystalline)
- 3) Gas sensing electrodes
- 4) Liquid membrane electrodes.
- 5) Enzyme based / Biochemical electrodes.

1 Glass membrane electrodes:

These electrodes are formed by doping molten SiO_2 with various chemicals this are H^+ sensitive the H^+ exchange with Li^+



diff Glass membranes are used like Na^+, Ag^+, K^+, NH_4^+

When membrane (glass) is dipped in solⁿ it

get swelled by forming hydrated gel layer so potential get developed in gel layer

$$Emf = E^0 + 0.0591 \log_{10} [M^{n+}]$$

↓
SE-P

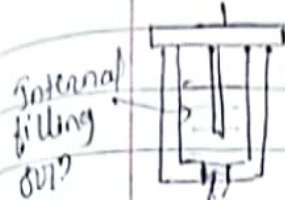
↓
conc of ions



Glass membrane electrode

2) Crystalline / Solid State membrane electrodes.

This membrane are made from ionic comp. or mix of ionic compound. They have good selectivity



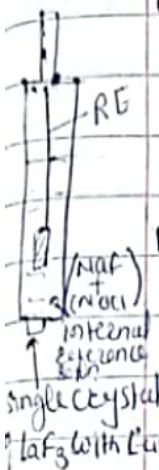
selectivity of crystalline membrane can be for both cation & anion as membrane forming material.

eg - 1) Fluoride ion determination.

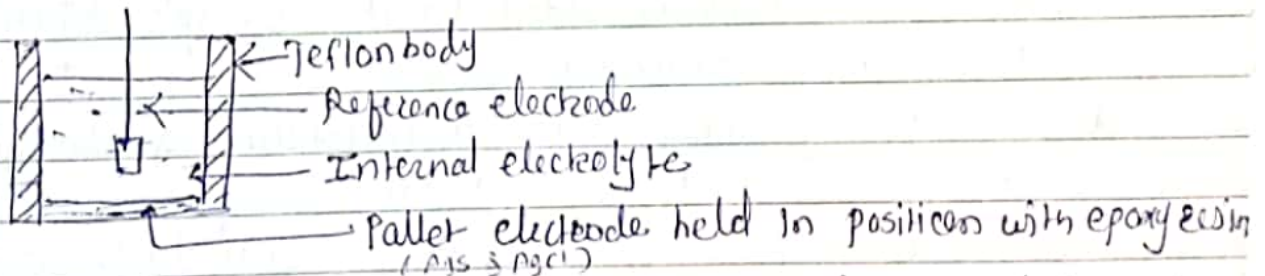
1) Homogeneous electrode of finely divided crystalline material like LaF_3 for determination of F^- in water.
 Lanthanum trifluoride

Construction

- 1) The membrane is made by doping europium fluoride (EuF_2) with Lanthanum trifluoride. The crystal is sealed at the bottom of polymeric container contain reference electrode. EuF_2 produce holes in crystal lattice of LaF_3 in which F^- ions can pass. When electrode dipped in soln. Electrode potential get developed across the membrane. Internal membrane is fixed potential depends on conc. of sample soln.



2) chloride ion determination



- Pallets of Ag_2S & AgCl is used. Electrode is made by Teflon. Pallet is attached using epoxy resin.
- Electrode potential get developed when electrode membrane contact with soln of Cl^- ions. This potential is measured against a constant reference electrode.

* Conductometry :

It is useful to measure electric conductance of solⁿ of like acids, bases & aq. salts. Electrolytes solⁿ due to presence of ions in solⁿ conduct electricity in presence of electric current.

- movement of ion depends on the mobility of the ion. This is depends on charge on ion, size & mass of ion & extent of solvation.
- Conductometry is used in neutralization titration, precipitation titration, chemical kinetics & plant laboratories etc.

1 Ohm's Law

Current (I) passing th^r a given solution is directly proportional to the potential diffⁿ betⁿ the two ends of the conductor th^r which the current is flowing.

$$I \propto E$$

$$\therefore E = RI \quad \left(\begin{array}{l} I - \text{Current strength,} \\ R = \text{proportionality const.} \\ \text{is Resistance.} \end{array} \right) \quad \left(E = \text{potential diffⁿ} \right)$$

$$\therefore R = \frac{E}{I}$$

Unit of R is Ω

2 Conductance (C)

Conductance is the reciprocal of resistance.

$$C = \frac{1}{R}$$

Unit of C = reciprocal of ohm or mho (ohm^{-1})

used in case current flows thr a conductor

3 Specific resistance:

Resistance of conductor at definite temp is directly proportional to its length & inversely proportional to area of cross section at const. temp.

$$R \propto \frac{l}{a} \quad \therefore \boxed{R = \rho \cdot \frac{l}{a}}$$

ρ = Specific resistance

Specific resistance of a sub? is defined as the resistance offered by a cube of unit length to the passage of electricity thr it.

$$\rho = R \cdot \frac{a}{l}$$



$$R \propto \rho$$

if $a=1, l=1$

$$\rho = \frac{\text{ohm} \cdot \text{cm}^2}{\text{cm}} = \text{ohm} \cdot \text{cm}$$

(SI unit ohm.m) ρ unit - ohm

4 Specific conductance:

Specific conductance of any conductor is reciprocal of specific resistance.

$$K = \frac{1}{\rho}$$

$$\text{sp resistance} = \frac{1}{K} = \frac{l}{a} = \frac{\text{cm}}{\text{ohm cm}^2} = \text{ohm}$$

Specific conductance of conductor - is the conductance of a cube of unit length.

Specific conductance of a solution - The conductance of solution betⁿ two parallel electrodes which have cross sectional area 1cm² which are kept 1cm apart.

$$\frac{1}{R} = \frac{1}{\rho} \cdot \frac{1}{l/a} \quad \therefore \frac{1}{R} = K = \frac{1}{l/a}$$

$$C = \frac{K}{l/a}$$

where C is conductance $K = C \cdot (l/a)$

$$K = \frac{\text{mho} \times \text{cm}}{\text{cm}^2} = \text{mho cm}^{-1}$$

Unit of specific conductance (K) = $\text{mho} \cdot \text{cm}^{-1}$

5 Cell constant (λ) - It is constant for given cell $\rightarrow$ ratio of (l/a) in the is called cell constant.

$$\text{Cell Const.} = \frac{\text{Specific Conductance}}{\text{measured conductance}} = \frac{l}{a} = \frac{\text{cm}}{\text{cm}^2}$$

Unit of cell const. = cm^{-1}

6 Equivalent Conductance :- measure of conducting power

The conductance of a solⁿ containing one gram equivalent of an electrolyte at any particular concentration, when placed betⁿ two electrode 1cm apart.

Equivalent conductance (λ) = sp. conductance $\times$ volⁿ in ml per gm equivalent of the electrolyte.

[If the conc. of solⁿ is c gm in equivalent per lit.

$$V = \frac{1000}{c} \quad \text{as } \lambda = K \times V$$

$$\lambda = \frac{K \times 1000}{c} = \frac{\text{ohm}^{-1} \text{cm}^{-1} \times \text{cm}^3}{\text{gm equiv}^{-1}}$$

Unit = $(\text{ohm}^{-1} \text{cm}^{-1}) \times \text{cm}^3 \cdot \text{gm}^{-1} \text{equiv}^{-1}$

$\therefore$ Unit of $\lambda = \text{ohm}^{-1} \cdot \text{cm}^2 \cdot \text{gm}^{-1} \text{equiv}^{-1}$

7 Molar conductance (μ) - Conductivity of the solⁿ containing 1 mole of the electrolyte. when placed betⁿ two large electrode, one cm apart (the conductance of all ions produced by ionization of 1 mole of an electrolyte when present in

$$\kappa \text{ (molar conductance)} = k \times V = k \times \frac{1000}{\text{molarity}}$$

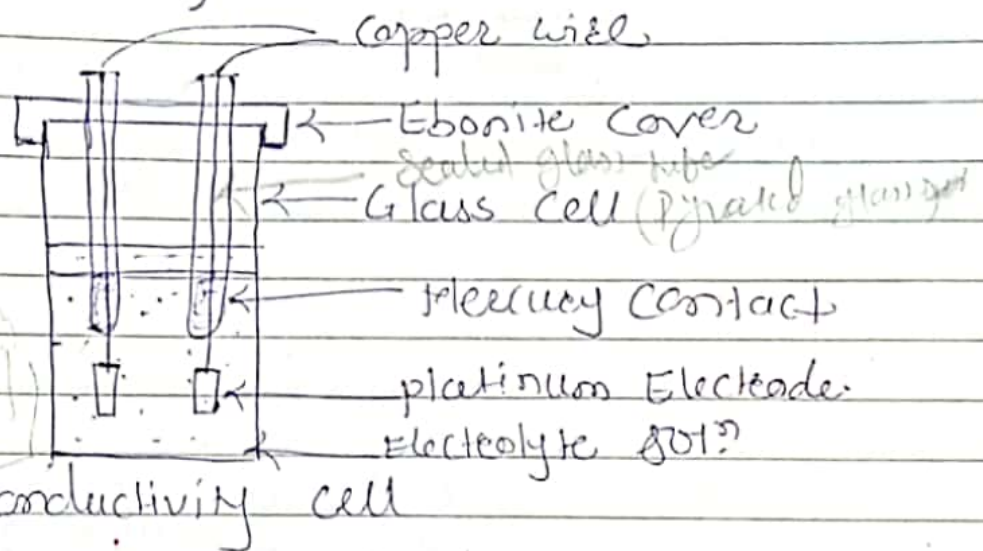
sp. conductance

$$\text{Unit of } \kappa = \text{ohm}^{-1} \text{cm}^{-1} \times \text{cm}^3 \text{mol}^{-1}$$

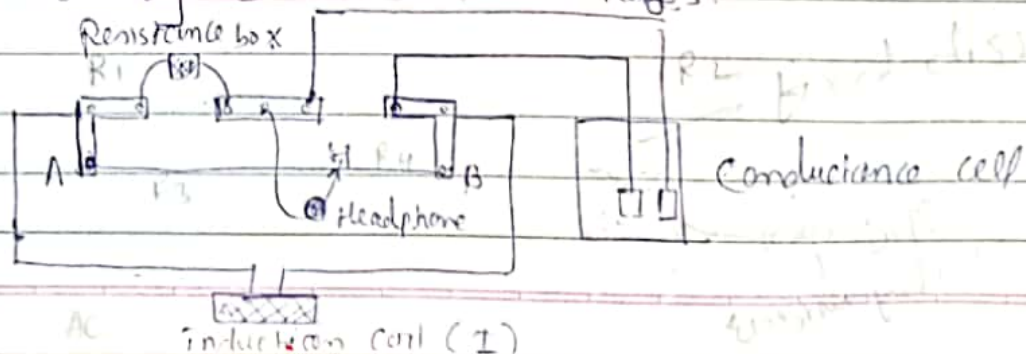
$$\kappa = \text{mho} \cdot \text{cm}^2 \cdot \text{g mol}^{-1}$$

* Conductivity cell:

- Conductance of solⁿ measured by using conductivity cell with alternate current.
- This can be done by Wheatstone bridge.
- The solution whose conductance is to be determined is placed in a special type of cell known as the conductivity cell.



- Electrodes are made by platinum plate coated with pt. black. pt. plates are welded to platinum wire which is fused in two glass tube. The contact with copper wires of the circuit is made by clipping them in mercury contained in the tubes.



→ DC - Current - electrolysis - resistance change

Galvanometer show no deflection of null pt.

- Instead of galvanometer a head phone is used. Manganese (Mn) wire is stretched over a meter AB. C is conductivity cell, R is resistance 50Ω . I is induction coil.

- When current flow from resistance is unplugged in the resistance box R. The sliding contact H is moved until the sound in the head phone is minimum. When this observe we have.

$$\frac{\text{Resistance of C}}{\text{Resistance of R}} = \frac{\text{Resistance BH}}{\text{Resistance AH}} = \frac{\text{Length BH}}{\text{Length AH}}$$

$$\therefore \text{Resistance of C} = \frac{\text{Length BH}}{\text{Length AH}} \times \text{Resistance R}$$

- Resistance of solⁿ in conductivity cell by eqⁿ convert to sp. conductance

$$K = \frac{1}{R} \times \frac{\text{dist. betⁿ two electrode}}{\text{Area of electrode}}$$

$$K = \frac{1}{R} \times \frac{l}{A}$$

$$K = \frac{1}{R} \times X$$

Where X is called as cell constant.

If specific conductance K ^{known then} equivalent conductance (A) & molar conductance (M) can be cal.

$$\Lambda = \frac{K \times 1000}{N}$$

(N is gm equivalent electrolyte)

$$\mu = \frac{K \times 1000}{M}$$

M = gm mole of electrolyte

* Determination of the cell cons.

If we know dist. betⁿ two electrodes & area of cross section we can cal. exact value of cell constant (l/a). ^{actual cell constant} measurement is difficult so indirect method is used.

$$\text{Specific conductance (K)} = \frac{1}{R} \times \frac{l}{a}$$

$$\text{Cell constant (X)} = \frac{\text{Specific conductance}}{\text{Observed conductance}} \quad \text{--- (1)}$$

- To determine cell cons. of KCl solⁿ if we know the specific conductance at given temp. Then KCl solⁿ of same concentration is prepared & its conductance is determined experimentally at same temp. then we can find cell constant by above formula. --- (1)

eg. Acc. to Kohlrausch law the sp. conductance of N/50 KCl solⁿ at 25°C is $0.002765 \text{ ohm}^{-1}$. Then in order to determine cell cons. N/50 KCl solⁿ is prepared by dissolving 0.372 gm of KCl in 250 ml water & conductance is determined at 25°C. The cell constant can be cal. by the observed conductance in the expression.

* Conductometric Titration :-

The titration in which end pt. of titration is noted by measurement of conductance of titration mix., known as Conductometric Titration.

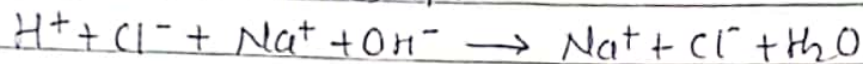
- Conductance of solⁿ at constant temp. depends on no. of ions present in it & their mobility.
- Titrant is added from burette into a measured vol^m of the solⁿ & various cellⁿ are plotted against vol^m of the titrant. Intersection

end point. — Acid-Base Titration

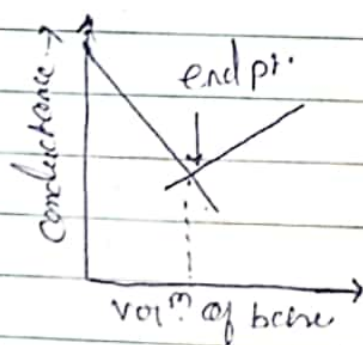
- 1) Strong Acid Vs Strong Base
- 2) Weak Acid Vs Strong Base
- 3) Weak Base Vs Strong Acid
- 4) Weak Acid Vs Weak Base

1 Titration of Strong acid Vs Strong Base

In the titration of strong acid like HCl against strong base NaOH from burette. The acid is taken in a beaker in which conductivity cell is immersed.



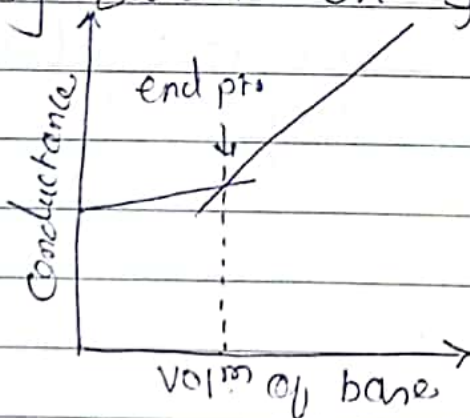
2. HCl is strong acid dissociate completely produce highly mobile H^+ & Cl^- ions. so conductivity at initially is high.
3. When NaOH is added from burette to HCl then H^+ ions are replaced by less mobile Na^+ ion. so conductivity decreases.
4. After complete neutralization. Na^+ & Cl^- ions concentration is increased & again conductivity is get increase due to presence of Na^+ & OH^- ions in the solution.
5. If Conductance is plotted against the vol^m of NaOH added.



2) Titration of weak acid vs Strong Base.

1. Titration of acetic acid (CH_3COOH) w. acid against a strong base (NaOH) (strong base)

$$\text{CH}_3\text{COOH} + \text{Na}^+ + \text{OH}^- \rightarrow \text{CH}_3\text{COO}^- + \text{Na}^+ + \text{H}_2\text{O}$$
2. Weak acid has poor low conductance because of the poor dissociation of weak acid.
3. On addition of NaOH from the burette the conductance increased due to increase amount of salt which dissociates completely.
4. At the end-pt. conductance is due to Na^+ & CH_3COO^- ions.
5. Further addition of NaOH conductance increases rapidly due to OH^- & Na^+ fast moving ions.

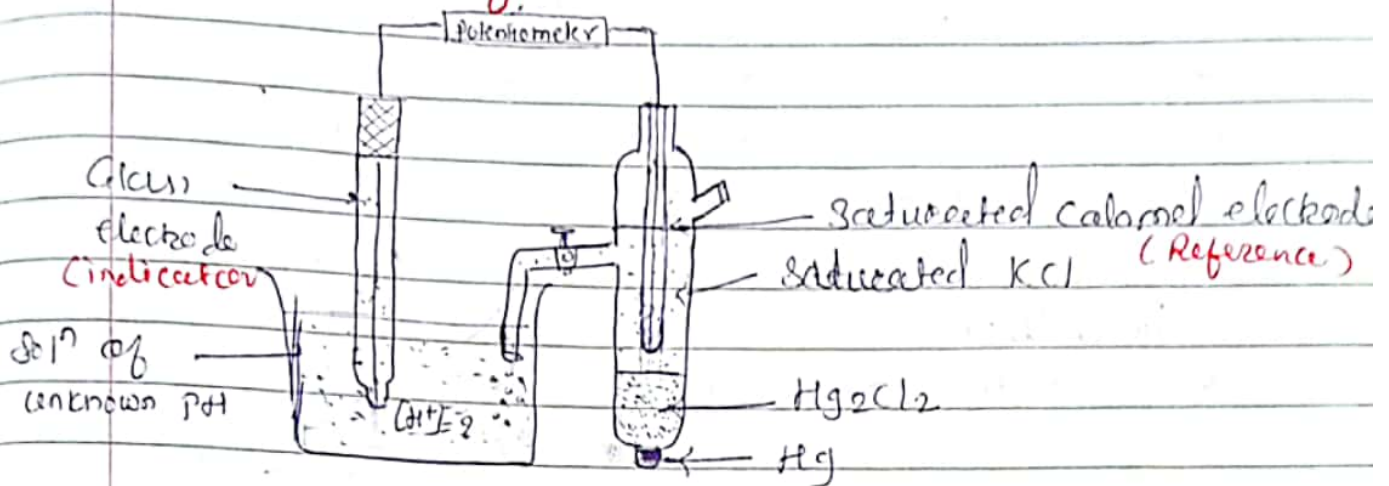


Titration curve weak acid vs S. Base.

* pH Metric Measurement:

In this method end point of acid-base titration can be determined.

1 Measurement of pH



1. Electrochemical cell consists of indicator electrode & reference electrode. These electrodes are immersed in the test soln & connected to a voltmeter.
2. pH is measured with the help of electrode potential.
3. The indicator electrode is glass electrode & for reference calomel electrode is used.
4. Glass electrode is sensitive to H⁺ ion so potential gets developed to the pH of the soln.

* Standardization (Calibration) of pH Meter:-

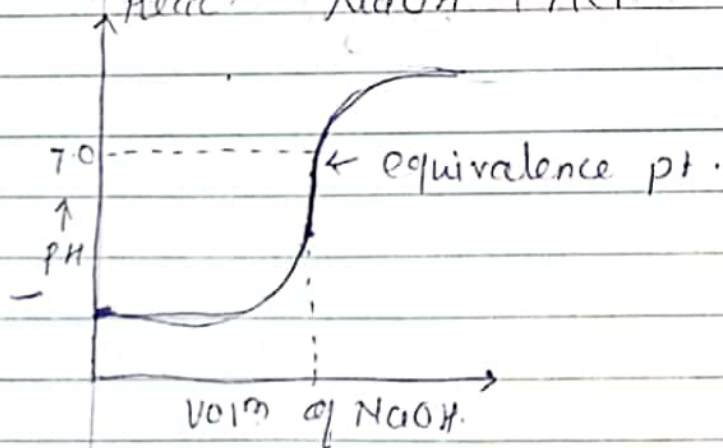
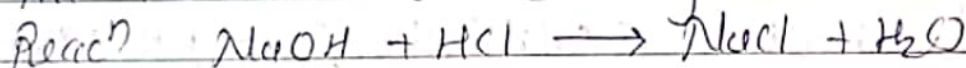
1. Buffer solution of pH 4 & 7 are used for standardization of pH meter.
2. Connect the reference & indicator electrodes to meter & switch on the pH meter.
3. First place electrode in the pH 7 solution for 1 min.
4. Then remove electrode & wash with distilled water.
5. Repeat these steps until a reliable measurement is obtained or adjust the pH 7.

* pH Metric Titration of Strong Acid against strong Base :-

1. Acid - Base titration is calculated by using pH meter.
2. S. Acid Vs S. Base when dissociate completely gives.
$$\text{Na}^+ + \text{OH}^- + \text{H}^+ + \text{Cl}^- \rightarrow \text{Na}^+ + \text{Cl}^- + \text{H}_2\text{O}$$
3. When Neutralization takes place at pH 7 means no. of H^+ & OH^- ions form water.
eg. Titration of HCl with 0.1N NaOH.

Procedure

- A Standardization of pH meter :-
By using Buffer solution of pH 4 and 7 pH meter is standardize.
- B Acid - Base titration
 - i. Take known quantity of HCl in a beaker.
 - ii. place the glass electrode (indicator) & Calomel electrode (Reference) in HCl solution.
 - iii. Filled the burette with 0.1N NaOH.
 - iv. Add 0.5 ml of NaOH to the Acid beaker & stirred the solⁿ note down the pH.
 - v. Continue addⁿ of 0.5 ml of NaOH & note the pH till the constant reading.



pH metric titration curve S. Acid Vs. S. Base

Observation:-

From the graph equivalent pt of titration is noted as 'x' ml. i.e. 0.1 N NaOH required for neutralization of HCl taken.

Calculation:-

$$N_1 V_1 (\text{HCl}) = N_2 V_2 (\text{NaOH})$$

$$N_1 = \frac{N_2 V_2}{V_1}$$

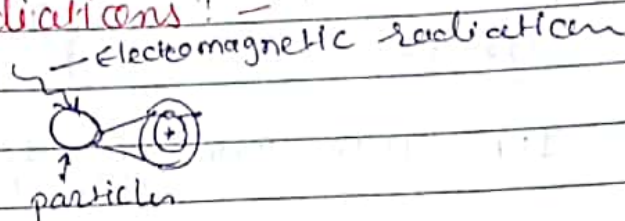
Normality of HCl can be calculated.

4 Spectroscopic Techniques

" Spectroscopy :-

It is a branch of science that deals with the study of interaction of electromagnetic radiations with matter.

Electromagnetic radiations :-

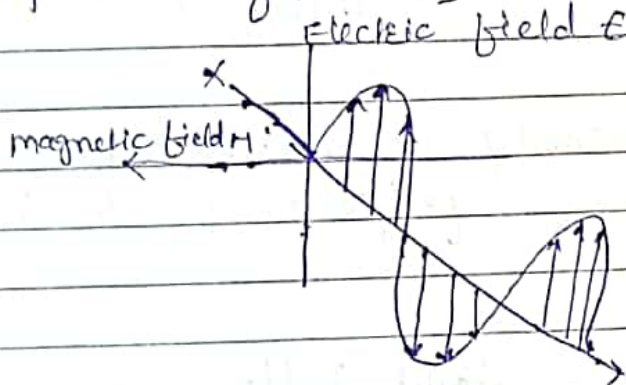


Electromagnetic radiation is a form of radiation energy like particles as well as wave nature.

→ Electric field

→ magnetic field.

"Electromagnetic radiations consist of discrete packets of energy are called photons." — Fundamental (subatomic particles that carry the electromagnetic force) i.e. light or particle (Photons are packet of energy)

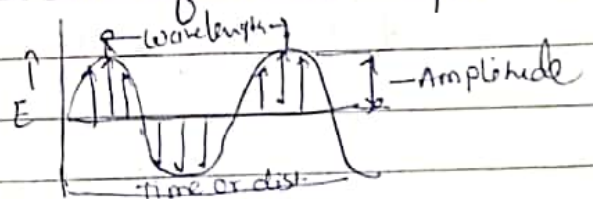


Direction of propagation.

Wavelength (λ) — distance between two crests or peaks (m)
freq. (ν) — The no. of waves that pass thr^r fixed pt. in unit time. (Hz)
wave number :- It is reciprocal of wavelength.

(cm^{-1})

$$E = h\nu = \frac{hc}{\lambda}$$



* UV-Visible Spectroscopy.

Absorption radiation in the UV - wavelength range $200 - 400 \text{ nm}$ & visible wavelength range $400 - 750 \text{ nm}$.
Ultra violet region subdivided into two parts.

① Near UV - $200 - 400 \text{ nm}$

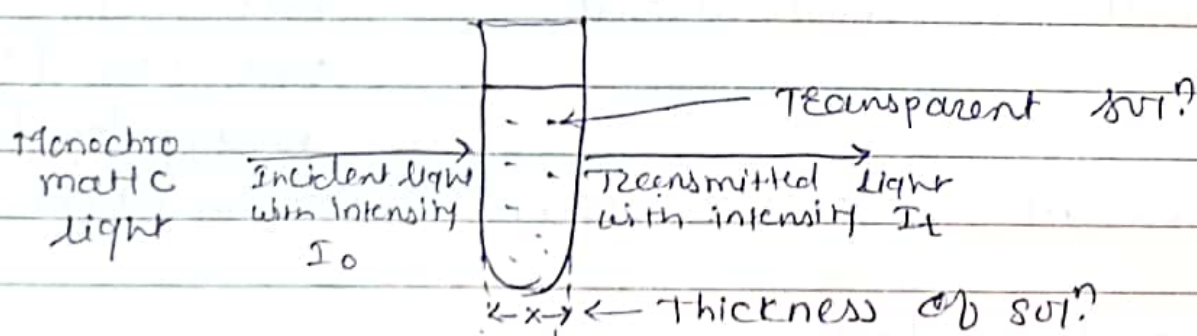
Far UV - below 200 nm

* Principle of UV-Visible Spectroscopy :-

"UV-visible spectroscopy is based on the absorption of ultraviolet light or visible light by chemical compounds, which results in the production of distinct spectra."

- Absorption spectra is useful in determination of structure of molecule."

* Beer's Law & Lambert's law.



When light is passed thr a sol? part of light is absorbed & remaining is transmitted.

$$A = I_0 - I_t \leftarrow \text{Transmitted light.}$$

$\uparrow$
Incident intensity

Beer's Law -
When a beam of monochromatic light is passed through a transparent solⁿ decrease in intensity of incident radiation is directly proportional to conc. at constant path length.

i.e. Absorbance is directly proportional to conc. (c) of solⁿ at constant path length.
 $\frac{dI}{I_0} \propto c \quad \text{--- (1)}$

(2) Lambert's Law -
When a beam of monochromatic light is passed through a transparent solⁿ decrease in intensity of incident radiation is directly proportional to path length at constant conc.
 $\frac{dI}{I_0} \propto dx \quad \text{or } A \propto x \quad \text{--- (2)}$
 [Absorbance is directly proportional to path length of solⁿ at constant conc.]

(3) Beer's-Lambert's Law.

" Absorbance (A) is directly proportional to path length (x) & concentration (c)

$$A \propto x \cdot c$$

$$\therefore A = a \cdot x \cdot c$$

a = proportionality constant called as absorptivity coefficient.

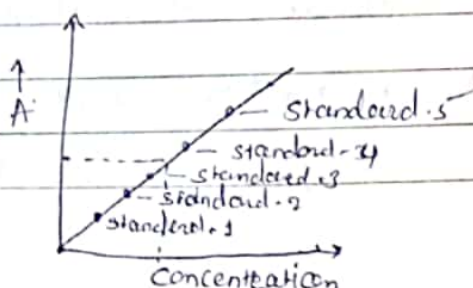
If $x = 1 \text{ cm}$, concentration $C = \text{mol/lit.}$ then ϵ

$\therefore$ Beer's-Lambert's law can be represented as

$$A = \epsilon \cdot x \cdot c$$

ϵ = molar absorptivity coefficient / molar extinction coefficient

* At constant path length absorbance vs concentration.



This law is used to cal. known concentration of sol?

Amount of light transmitted is given by transmittance (T)

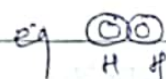
$$T = \frac{I_t}{I_0}$$

Rel. Rel? $\Delta \log T$
 $A = -\log T$

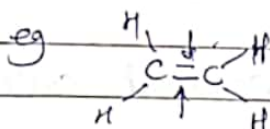
* Absorption of UV Radiation by organic molecule leading to different Electronic Transition.

Organic molecule

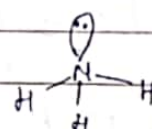
σe^-
 e^- forming single bonds are called σ
 e^- formed in Saturated Hydrocarbons C-H bonds



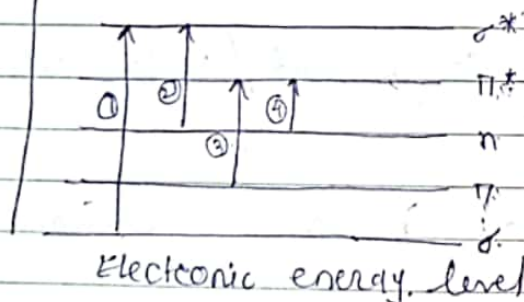
πe^-
 e^- in forming double bonds are called πe^- . formed in unsaturated hydrocarbons C=C, C $\equiv$ C



$n e^-$
 These are non bonded or lone pair of e^- not involving in bonding.
 eg - Oxygen, Nitrogen



Energy.



Possible e^- transitions are (1) $\sigma \rightarrow \sigma^*$ (2) $n \rightarrow \sigma^*$ (3) $\pi \rightarrow \pi^*$ (4) $n \rightarrow \pi^*$
 $\sigma \rightarrow \pi^*$ & $\pi \rightarrow \sigma^*$ are forbidden transitions.

$$\sigma \rightarrow \sigma^* > \sigma \rightarrow \pi^* \sim \pi \rightarrow \sigma^* > \pi \rightarrow \pi^* \sim n \rightarrow \sigma^* > n \rightarrow \pi^*$$

The relative energy are in the given order as above.

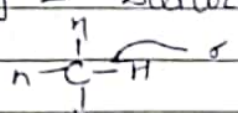
Selection rule $\sigma \rightarrow \sigma^*$ & different orientation
② energy diff. ③ selection rule
 σ low an π more.



When excitation occurs e^- from one of the filled σ, π or n molecular ^(bonding) orbital to vacant σ^* or π^* (antibonding) molecular orbitals.

① $\sigma \rightarrow \sigma^*$ transition:-

The excitation of e^- from bonding to antibonding is $\sigma \rightarrow \sigma^*$. The e^- s are from σ bonds are tightly held & hence reqd large energy for excitation & absorption.

eg - Saturated hydrocarbons. methane CH4
 So $\sigma \rightarrow \sigma^*$ transition takes place.

In methane Absorbance maxima at 125 nm

* Absorbance takes place at short wavelength in the range 120 - 200 nm. It is not seen in range (200 - 800 nm)

2) $\pi \rightarrow \pi^*$

The excitation of e^- from bonding π to higher energy antibonding π^* orbital. This transition shows in comp. containing multiple bonds like alkene, alkynes, aromatic comp.

- Alkene absorb in the region 170 - 205 nm in far UV region so it difficult to record by Spectrophotometer. If comp. absorbed more than 200 nm can be recorded easily.

3) $n \rightarrow \sigma^*$

excitation of e^- from non bonding orbital to higher energy antibonding σ^* orbital. Saturated compounds contain heteroatoms like O, N, S & halogens. This reqd less energy & hence occurs at longer wavelength. eg - Alcohol, Amines.

4) $n \rightarrow \pi^*$

The excitation of e^- from non bonding atomic orbital to higher energy antibonding π^* orbital. Comp. contain double bond with hetero atoms eg ($C=O$, $C=S$, $N=O$ etc)

The $n \rightarrow \pi^*$ transition req^d less energy of absorption takes place at longer wavelength. eg - Acetone. Show absorption bands one at 190 nm & other at 280 nm.

* Absorption at 280 nm is due to $n \rightarrow \pi^*$ transition. req^d less energy.

* Absorption at λ 190 nm is due to $\pi \rightarrow \pi^*$ which req^d higher energy.

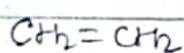
* Imp. Terms in UV-Visible Spectroscopy.

1) Chromophore -

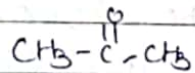
Colour bearing

" Chromophore is a covalently unsaturated group responsible for electronic absorption in UV region. due to $\pi \rightarrow \pi^*$ transition & $n \rightarrow \pi^*$ transitions at absorption radiation eg - above 200 nm.

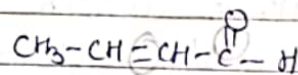
eg - $C=C$, $C=N$, $C=O$, $N=N$ etc.



$\lambda_{max} = 171 \text{ nm}$



$\lambda_{max} = 279 \text{ nm}$



$\lambda_{max} = 290 \text{ nm}$

Conjugation λ_{max} to longer λ

② Auxochrome -

It is saturated functional gr. which itself does not absorb radiation in near UV region, but when attached to chromophore shift the absorption to longer wavelength.

eg - OH , $-OCH_3$, $-NH_2$ etc



$$\lambda_{\text{max}} = 255 \text{ nm}$$



auxochrome

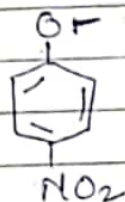
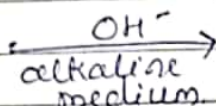
chromophore

$$\lambda_{\text{max}} = 270 \text{ nm}$$

3) Bathochromic Shift / Red shift:

A shift of absorption to longer wavelength is known as bathochromic shift.

eg - p nitrophenol shows red shift in alkaline medium.



$$\lambda_{\text{max}} = 255 \text{ nm}$$

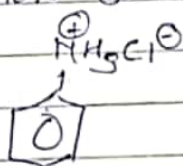
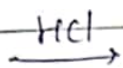
$$\lambda_{\text{max}} = 265 \text{ nm}$$

This effect takes place due to presence of auxochrome or by change the solvent.

4) Hypsochromic shift / Blue shift:

A shift of absorption to shorter wavelength is known as Hypsochromic shift or blue shift.

eg - Aniline show blue shift in acidic cond?



$$\lambda_{\text{max}} = 285 \text{ nm}$$

$$\lambda_{\text{max}} = 265 \text{ nm}$$

5) Hyperchromic shift :-

An increase in intensity of absorption maxima is called hyperchromic shift.



$$\lambda_{\text{max}} = 257 \text{ nm}$$

$E_{\text{c}} = 2750$
(Absorption intensity)

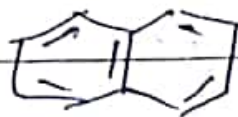


2-methyl pyridine
 $\lambda_{\text{max}} = 260 \text{ nm}$
 $E_{\text{c}} = 3560$

6) Hypochromic Shift: -

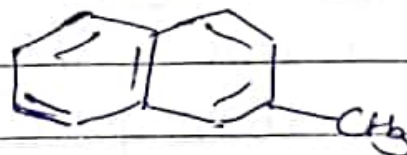
Decrease in intensity of absorption maxima is called hypochromic shift.

It occurs due to introduction of gr. that distorts the original geometry of molecule.



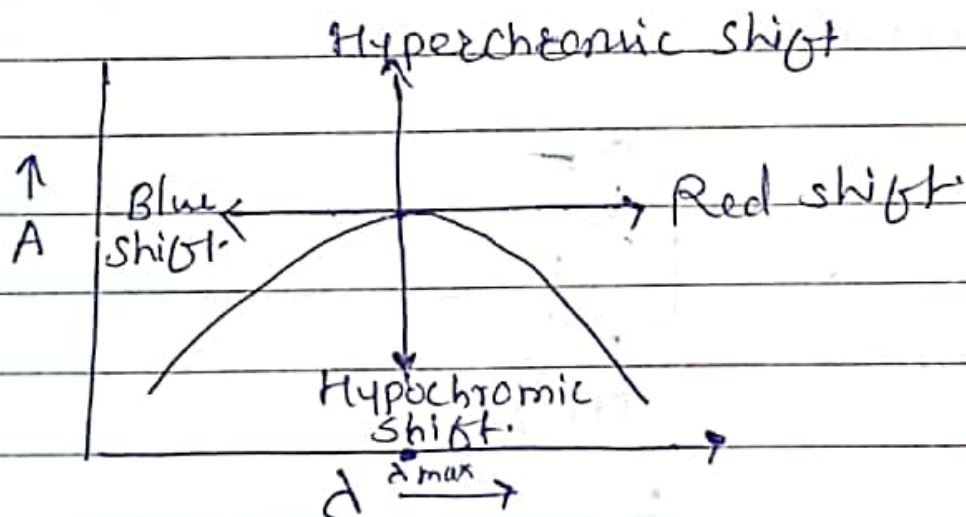
Naphthalene

$$\epsilon = 19000$$



2-methyl naphthalene

$$\epsilon = 10250$$



* Applications of UV-visible Spectroscopy! -

1) Qualitative analysis:

i) Detection of functional group:-

The technique is used to determine presence or absence of chromophore. It can be helpful to determine absence of conjugation, $C=O$ or Aromatic Comp.

ii) Extent of conjugation:-

If double bonds get increases it shows longer wavelength.

iii) Distinguish conjugated & nonconjugated double bonds.

- Conjugated shows longer wavelength than nonconjugated.

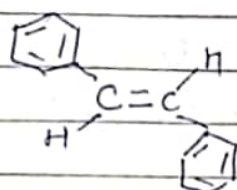
iv) Identification of unknown comp.

- Absorbance value help to identify unk. comp.

2) Determine of cis & trans comp.:-

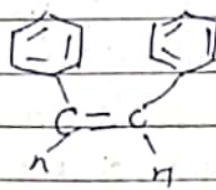
eg. It is useful to find out geometrical isomers.

eg



trans - stilbene

$\lambda_{max} = 295 \text{ nm}$



cis stilbene

$\lambda_{max} = 280 \text{ nm}$

Trans isomer shows more λ_{max} due to less steric hindrance but in cis due to steric hindrance λ_{max} get decreases.

3) Determine mol. wt. :-

Using Lambert's - Beer's law i.e. $A = \epsilon \times C \times l$, mol. wt. of comp. can be determined.

4) Detection of impurities:-

Some additional peak can be observed if impurities are present. By measuring absorbance at particular wavelength, impurities can be determined.

5) Quantitative analysis -

Quantitative analysis takes place by Lambert's - Beer's law

6) Study of kinetics of chemical reactions